# Shuyi (Ryan) Chen

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CONTACT

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EDUCATION

**Carnegie Mellon University**, Heinz College — Pittsburgh, PA
Ph.D. in Information Systems Management — 2024 – Aug 2027 (*expected*)
Advisors: Prof. Shixiang (Woody) Zhu and Prof. Ramteen Sioshansi
Master of Information Systems Management (Highest Distinction) — 2021 – 2023

**The Chinese University of Hong Kong, Shenzhen** — Shenzhen, China
B.B.A. (Hons.) in Global Supply Chain and Logistics Management; Minor in Statistics — 2017 – 2021
First Class Honours; CUHK Presidential Award
Visiting Student, University of California, Berkeley — 2018

RESEARCH INTERESTS

- Power-system resilience application: sectionalization and Public Safety Power Shutoffs; outage management; energy storage operations
- Learning and optimization under uncertainty: decision-focused learning; robust and chance-constrained optimization; conformal prediction; generative models

PUBLICATIONS

JOURNAL ARTICLES

[J1] **Global-Decision-Focused Neural ODEs for Proactive Grid Resilience Management** [arXiv] [code]
Shuyi Chen, Ferdinando Fioretto, Feng Qiu, Shixiang Zhu.
*IEEE Transactions on Smart Grid.*

REFEREED CONFERENCE PROCEEDINGS (*Equal contribution)

[C1] **Learning Polyhedral Conformal Sets for Contextual Robust Optimization** [arXiv]
Shuyi Chen*, Wenbin Zhou*, Shixiang Zhu.
*Conference on Neural Information Processing Systems (NeurIPS), 2026.*

[C2] **Gen-DFL: Decision-Focused Generative Learning for Robust Decision Making** [arXiv] [code]
Prince Zizhuang Wang, Shuyi Chen, Jinhao Liang, Ferdinando Fioretto, Shixiang Zhu.
*International Conference on Learning Representations (ICLR), 2026.*
IISE DAIS Track Best Paper Award.

[C3] **Counterfactual Fairness through Transforming Data Orthogonal to Bias** [arXiv] [code]
Shuyi Chen, Shixiang Zhu.
*ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), Research Track, 2025.*

[C4] **Uncertainty-Aware Robust Learning on Noisy Graphs** [arXiv] [poster] [code]
Shuyi Chen, Kaize Ding, Shixiang Zhu.
*IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2025.*
Short version at the NeurIPS 2023 Workshop on New Frontiers in Graph Learning.

[C5] **API-Assisted Code Generation for Question Answering on Varied Table Structures** [arXiv]
Yihan Cao*, Shuyi Chen*, Ryan Liu*, Zhiruo Wang, Daniel Fried.
*Conference on Empirical Methods in Natural Language Processing (EMNLP), 2023.*

## Working Papers

### Under Revision

[W1] **Enhancing Electricity-System Resilience with Adaptive Robust Optimization and Conformal Uncertainty Characterization** [arXiv] [media]
Shuyi Chen, Shixiang Zhu, Ramteen Sioshansi.
*Minor revision, IEEE Open Access Journal of Power and Energy.*
INFORMS ENRE Best Student Paper Award.

### In Preparation

[W2] **Large-Scale Resilience Planning for Wildfire-Prone Electricity Systems via Adaptive Robust Optimization** [arXiv] [poster] [demo]
Shuyi Chen, Shixiang Zhu, Ramteen Sioshansi.
*To be submitted.*

[W3] **The Price of Looking Ahead: Decision-Focused Learning for Model Predictive Control in Energy Storage**
Shuyi Chen, Shuai Wang, Shixiang Zhu.
*In preparation for conference submission.* Earlier version: "Balancing Planning Myopia and Forecast Error: Decision-Focused Sequential Learning."
INFORMS DMDA Workshop Best Paper Finalist (Applied Track).

[W4] **Rare-Chance-Constrained Resilience Planning for Power Systems under Wildfires and Cascading Hazards**
Shuyi Chen, Shixiang Zhu, Ramteen Sioshansi.
*In preparation for journal submission.*

## Honors & Awards

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|---|---|
| INFORMS DMDA Workshop Best Paper Finalist (Applied Track) | 2026 |
| IISE DAIS Track Best Paper Award | 2026 |
| INFORMS ENRE Best Student Paper Award | 2025 |
| KDD Student Travel Award; NSF Travel Grant (IMSI Workshop) | 2025 |
| INFORMS Workshop on Data Science Student Scholarship | 2024 |
| PwC Technology Impact Office / CMU Digital Transformation Center Research Fellowship | 2023 |
| CUHK Presidential Award (top 10 of 1,200 graduates) | 2021 |

## Teaching

**Lecturer**, *A/B Testing* (95-819), Heinz College, Carnegie Mellon University 2026
**Teaching interests:** optimization and operations research; machine learning and data analytics; A/B testing and causal inference; energy and sustainability analytics.

## Presentations

INFORMS Annual Meeting, San Francisco (upcoming) 2026
*Large-scale resilience planning for wildfire-prone electricity systems*
Young Researchers Workshop, Cornell University 2026
*Rare-chance-constrained resilience planning under wildfires and cascading hazards*
HOPE Research Workshop, University of Notre Dame (invited talk) 2026
*Large-scale resilience planning for wildfire-prone electricity systems*
ACM SIGKDD Conference, Toronto 2025
*Counterfactual fairness through transforming data orthogonal to bias*
NSF/IMSI Workshop on UQ and ML for Complex Physical Systems, Chicago (poster) 2025
*Electricity-system resilience with adaptive robust optimization and conformal uncertainty*
INFORMS Workshops on Data Science and on Data Mining & Decision Analytics 2024
*Counterfactual fairness*; *decision-focused neural ODEs for grid resilience*

## Industry Experience

**Public Advocates Office, California Public Utilities Commission** — San Francisco, CA
Visiting Scholar, hosted by Dr. Iain Fisher — Oct 2026 – present

• Work with the office on wildfire resilience planning [W2] and on undergrounding decisions under rare, high-consequence hazards, motivated by California's Senate Bill 884 expedited undergrounding program.

**Argonne National Laboratory** — Lemont, IL
Researcher, hosted by Dr. Feng Qiu — 2025

• Developed decision-focused neural ODEs for proactive grid resilience that model how outages spread over space and time, reducing decision regret by up to 75% versus two-stage baselines [J1].

• Led the 2025 INFORMS DMDA Workshop Data Challenge on extreme-weather power-outage prediction, which drew 54 teams from 31 institutions.

**ByteDance** — Beijing, China
Data Scientist — 2023 – 2024

• Built Prophet-based anomaly detection with statistical residual tests (3% MAPE), flagging three major business anomalies for timely mitigation.

• Applied causal inference to Local Services growth interventions, contributing to a 2% increase in overall user activity.

## Public Platform

• Led development of a wildfire analytics platform (2026) for wildfire-mitigation policy: FastAPI services on AWS over a PostGIS warehouse of California's wildfire, shutoff, and outage records; point-process ignition-risk hindcasts; optimization-based planning studies; and a tool-calling LLM agent whose harness validates every call and cites the query behind every number. [link]

## Professional Service

**Competition Co-Chair:** INFORMS DMDA Workshop Data Challenge (2025); IISE ESD Energy Analytics Challenge (2026).

**Organizing Committee:** YinzOR, Speakers and Talks Committee (2026).

**Session Chair:** INFORMS Energy Systems session (2025).

**Journal Reviewer:** INFORMS Journal on Data Science; Journal of Agricultural, Biological, and Environmental Statistics.

**Conference Reviewer:** NeurIPS; ICML; ICLR; KDD; AISTATS; ICASSP; IEEE PES General Meeting (2023 – 2026).

## References

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Principal Computational Scientist
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**Ferdinando (Nando) Fioretto**
Associate Professor, Computer Science
University of Virginia
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